Strategic Investment with Asymmetric Information, Jeffrey Updating, and Endogenous Volatility

A Self-Contained Model with Drift-Risk Decoupling

Contents

1 Overview and Motivation

This document presents a complete, self-contained model of strategic investment between two asymmetrically informed players. Each player owns a risky stochastic process and may invest in the other's process for a fee. Players market themselves to attract investment, and beliefs are updated via Jeffrey conditioning applied to a Beta-Binomial model of returns.

The model features a clean microeconomic mechanism for endogenous volatility: when player i receives investment from player j, the incoming capital multiplicatively augments the growth (drift) base of i's process without increasing the volatility borne by i. Player i bears risk only on their own retained capital C_i, while capturing fee income on the larger asset base. This decoupling of growth benefits from risk costs generates rich strategic feedback into realised portfolio volatility through Jeffrey belief updating and marketing.

The key outputs are:

(i) Equilibrium allocations α_i^* and marketing spends m_i^* in closed form.

(ii) Realised portfolio variance derived from the strategic interaction.

(iii) A structural GARCH(1,1) representation with endogenous GJR-type asymmetry.

All objects are defined explicitly to permit full symbolic solution in SymPy.

2 Primitives and Notation

2.1 Players and Budgets

There are two players, indexed $i \in \{1, 2\}$, with $j = 3 - i$.

Definition 2.1 (Budgets). *Player i begins with initial wealth $W_i > 0$ and chooses marketing spend $m_i \in [0, W_i]$, leaving effective wealth*

$$W_i^{\text{eff}} = W_i - m_i.$$

2.2 Return Processes

Each period, player i's process generates return $r_i^t = u_i$ with true probability p_i (known only to i) or d_i otherwise, where $u_i > d_i$. The true mean and variance are

$$\mu_i = p_i u_i + (1 - p_i) d_i, \qquad \sigma_i^2 = p_i (1 - p_i)(u_i - d_i)^2.$$

2.3 Public History and Beliefs

Both players observe the same history of n returns with k_i ups. The Bayesian posterior is Beta($\alpha_i^* = \alpha_i^0 + k_i, \beta_i^* = \beta_i^0 + n - k_i$). Marketing by i generates a Jeffrey shift $\delta(m_i) = m_i/(m_i + \gamma_i)$, yielding Jeffrey-shifted parameters $\alpha_i^{**} = \alpha_i^* + \delta(m_i)$, $\beta_i^{**} = \beta_i^*$ and perceived probability $\hat{p}_i^j = \alpha_i^{**}/(\alpha_i^{**} + \beta_i^{**})$.

Perceived moments for player j about i's process are

$$\mu_i^j = \hat{p}_i^j u_i + (1 - \hat{p}_i^j) d_i, \qquad \sigma_i^{j2} = (u_i - d_i)^2 \hat{p}_i^j (1 - \hat{p}_i^j).$$

2.4 Player's Investment Choices

Each player has exactly three uses for their capital and no others:

1. **Own process** (Choice 1): Retain fraction $(1 - \alpha_i)$ of W_i^{eff} in own process. This capital alone bears true volatility σ_i^2.
2. **Marketing** (Choice 2): Spend m_i to shift opponent's beliefs upward, attracting more incoming capital without changing true μ_i or σ_i^2.
3. **Opponent's process** (Choice 3): Allocate fraction α_i of W_i^{eff} to j's process, receiving net-of-fee return based on own Jeffrey beliefs.

Incoming investment is accepted passively.

2.5 Why Jeffrey Conditioning?

Standard Bayesian updating requires *hard evidence* — an observation that becomes certain once received. Jeffrey conditioning, by contrast, is designed for *soft evidence*: the agent revises their degree of belief without any event being assigned probability one. This distinction is essential for modelling marketing in the present framework.

Table 1: Why the Model Uses Jeffrey Conditioning

Aspect	Standard Bayesian Updating	Jeffrey Conditio
Signal type the method can handle	Hard, verifiable observations only	Uncertain / soft
Update rule	Conditions on an event with probability 1	Revises credence
Treatment of marketing	Would require verifiable disclosure or audit	Naturally models
Effect on perceived volatility	Almost always reduces uncertainty	Can raise or lowe
Economic interpretation	"Revealing objective information"	"Marketing a lie
Key rigidity preserved	—	Conditional likel
Role in volatility dynamics	Limited strategic feedback	Enables full endo

Jeffrey conditioning therefore provides the precise mechanism needed: it allows marketing to be a *costly, dissipative, soft-evidence signal* that manipulates the opponent's perceived drift and volatility without altering the true stochastic process. This generates the rich endogenous volatility dynamics that are the central contribution of the model.

Definition 2.2 (Jeffrey Conditioning — Mathematical Derivation)**.** *Let H be the hypothesis of interest (here: "the true up-probability is p_i") and let the evidence partition the space into $\{E, \neg E\}$, where E represents "favourable evidence for high p_i".*

Jeffrey's rule of conditioning (Jeffrey, 1965) states that if the agent revises their credence in the evidence from the prior $P(E)$ to a new value $q \in (0, 1)$, the updated probability of H is

$$P'(H) = P(H \mid E) \cdot q + P(H \mid \neg E) \cdot (1 - q).$$

No event is treated as certain; only the weight on E changes.

In the Beta-Binomial setting of this model: - The prior/posterior after public history is $\mathrm{Beta}(\alpha_i^*, \beta_i^*)$. *- The mean is $\hat{p}_i^j = \frac{\alpha_i^*}{\alpha_i^*+\beta_i^*}$.*

*Marketing supplies soft evidence equivalent to raising the credence in "up" by the amount $\delta(m_i) \in (0, 1)$. This is mathematically identical to adding a fractional pseudo-count $\delta(m_i)$ **only to the up-count** while leaving the down-count unchanged:*

$$\alpha_i^{**} = \alpha_i^* + \delta(m_i), \qquad \beta_i^{**} = \beta_i^*.$$

The new perceived up-probability becomes

$$\hat{p}_i^j = \frac{\alpha_i^{**}}{\alpha_i^{**} + \beta_i^{**}} = \frac{\alpha_i^* + \delta(m_i)}{\alpha_i^* + \delta(m_i) + \beta_i^*}.$$

To see that this is exactly Jeffrey's rule, note that the Beta mean is a weighted average of the conditional means under E and $\neg E$. Adding $\delta(m_i)$ to α_i^* is equivalent to setting $q = \frac{\alpha_i^*+\delta(m_i)}{\alpha_i^*+\beta_i^*+\delta(m_i)}$ in the Jeffrey formula above, confirming that the update is a convex combination that shifts credence without treating any new observation as certain.

This derivation holds because the conditional likelihood $P(r^t = u_i \mid p_i) = p_i$ is unaffected by marketing (rigidity condition).

3 Terminal Wealth and Preferences

Definition 3.1 (Exposures)**.** *Define*

$$C_i = (1-\alpha_i)W_i^{\text{eff}} \qquad \textit{(own retained capital that bears risk)}, \tag{1}$$
$$A_i = C_i + f\alpha_j W_j^{\text{eff}} \qquad \textit{(drift/growth exposure)}, \tag{2}$$
$$B_i = \alpha_i W_i^{\text{eff}}(1-f) \qquad \textit{(risk exposure to opponent's process)}. \tag{3}$$

Terminal wealth is

$$W_i(T) = \mu_i A_i + \mu_j^i B_i + C_i \cdot \varepsilon_i + B_i \cdot \tilde{\varepsilon}_j.$$

Mean-variance utility:

$$\mathcal{V}_i = \mathbb{E}[W_i(T)] - \frac{\rho_i}{2}\mathrm{Var}[W_i(T)],$$

with

$$\mathbb{E}[W_i(T)] = \mu_i A_i + \mu_j^i B_i, \qquad \mathrm{Var}[W_i(T)] = C_i^2\sigma_i^2 + B_i^2\sigma_j^{i2}.$$

4 Stackelberg Equilibrium

Player 1 (leader) moves first choosing (m_1, α_1); player 2 (follower) responds with $(m_2 = 0, \alpha_2)$.

Useful shorthand:

$$\Delta_i = \mu_j^i(1-f) - \mu_i, \quad \Sigma_i = \sigma_i^2 + (1-f)^2\sigma_j^{i2}.$$

Optimal allocations:

$$\boxed{\alpha_2^* = \frac{\Delta_2 + \rho_2\sigma_2^2 C_2}{\rho_2 W_2^{\text{eff}}\Sigma_2}}, \qquad \boxed{\alpha_1^* = \frac{\Delta_1 + \rho_1\sigma_1^2 C_1}{\rho_1 W_1^{\text{eff}}\Sigma_1}}.$$

The leader's optimal marketing m_1^* retains the square-root closed form derived from $\partial\mathcal{V}_1/\partial m_1 = 0$, with the new variance plugged in (explicit expression obtainable symbolically).

Follower sets $m_2^* = 0$.

5 Realised Portfolio Volatility

At equilibrium,

$$\mathcal{R}_i = C_i^2 \sigma_i^2 + B_i^2 \sigma_j^{i2},$$

evaluated at $(m_1^*, \alpha_1^*, \alpha_2^*)$. Marketing tends to reduce $\mathcal{R}_i$ via the protected risk exposure C_i.

6 Structural GARCH Representation

[Insert the full updated GARCH section I provided in my previous message here — the long block starting with "Multi-Period Extension" and ending with the stationarity remark.]

7 Complete Parameter List

[Keep your original table, adding rows for C_i and A_i under equilibrium objects.]

8 Guide to Symbolic Solution in SymPy

[Keep or slightly update your original guide; the structure remains the same with the addition of $C_i = (1 - \alpha_i) W_i^{\text{eff}}$.]